17. A particle of mass 2 kg moves under a force $F = 3x$ N. Find its acceleration at $x = 4$ m.

18. Two blocks of mass 1 kg and 2 kg are connected by a string over a frictionless pulley mounted at the edge of a table, with the 2 kg block hanging vertically and the 1 kg block resting on the horizontal table. Assuming the table and pulley are frictionless and the string is inextensible, determine the tension in the string connecting the two blocks

and the resulting linear acceleration of the system once released from rest.

19. A car accelerates uniformly from rest to 20 m/s in 5 seconds. Calculate the distance covered.